
Paper 019a: Observe Before Optimizing

An Attribution Addendum for Packer-Sourced FA3 Boundaries

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Abstract

This prospective addendum strengthens only the Round-1 execution contract of paper 019 before any governed performance diagnostic. The candidate moves document-boundary discovery from a per-step CUDA scan to exact metadata already known by the packer. A newer independent critic lowers its local impact and transfer scores, from the paper-canonical 33/40 to 30/40, while retaining it as a diagnostic-first dependency. We freeze a 1,000-batch pre-clock equivalence gate, a sequential 20-step scanner/scanner A/A phase, a 20-step scanner/sidecar A/B phase, a 10-step washout, and exactly 200 clean timing samples. Model, optimizer, data, boundary, loss, CPU-RNG, CUDA-RNG, and compiler identities are bound at the relevant phase boundaries. Two independent H200 executions already pass adversarial event-lifetime, dynamic shape, and real FlashAttention-3 forward/backward tests; that evidence validates the implementation substrate but is neither a speed result nor a RunRecord. A single physical placement can only kill the mechanism or require a separately preregistered role-swapped profile. It cannot authorize a validation-BPB endpoint. The historical 0.927183 remains the chart origin and a quarantined provenance observation; this addendum reports no new SOTA.

1. Authority, Scope, and Non-Claims

Paper 019 preregistered a connected five-round program (GPT Research Coordinator, 2026). Its im-

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Internal OPHIS preregistration in ICML two-column format. Not peer reviewed, submitted to, or accepted by ICML.

Table 1. Explicit ratings frozen before the governed diagnostic.

assessment	N/P/V/I/R/F/X/C	sum
original proposer	4/4/5/3/4/4/5/5	34
paper-019 critic	4/4/5/3/3/4/5/5	33
new independent critic	4/4/5/2/2/3/5/5	30

mutable TeX and PDF SHA-256 values are respectively 9ba9d7bcd7a29612bf2bc4fb6d7b8cfe69c53969187793e8f5b7960de164bc5 and 17a8e9d3e2af669bb4b0c105c1ec7965ad0cd9f8bc59fc7467230526db839db8. This append-only addendum supersedes its Round-1 execution details where the two disagree; it does not change the other four rounds, alter their order, or authorize an experiment. In particular:

no scored endpoint, no validation BPB,
no comparator rebind, and no new SOTA.

The chart must continue to begin at historical experiment 501, its original timestamp, and val_bpb = 0.927183. That point is not a contemporaneous control. The active causal question remains sidecar minus scanner under the same recovered recipe and a fresh governed seed.

This document was frozen after implementation-level systems validation but before the governed phased diagnostic. Preflight failures and fixes are preserved rather than erased. A diagnostic may execute only after typed registration, setup reconciliation, source-hash matching, strict repository audit, and a fresh tenant-free GPU inventory.

2. Explicit Ratings and Selection Rationale

Every vector uses the fixed order novelty (N), primary-source provenance (P), causal validity (V), expected local impact (I), transfer reliability (R), feasibility/cost (F), falsifiability (X), and connected-program coherence (C). All dimensions are scored from 1 to 5, where higher is better.

The lower live score is authoritative for current risk

assessment. The candidate is not expected to improve modeling quality directly, and the cost removed may be too small relative to GPU-placement variance. It remains first for a dependency reason, not because 30/40 is the largest portfolio score: exact boundary metadata is needed to observe and attribute later document-local mechanisms, and an accepted systems gain would increase the charged update exposure of every downstream round. This preserves the campaign principle:

Before a system can improve itself, it must be able to observe and attribute its failures.

The strongest model reserve proposed after the plateau is GPAS residual activation control, proposer score $4/5/5/4/4/5/5/5 = 37/40$ (Chen et al., 2025). The strongest independently criticized Engram child is distinct prime moduli per hash head/layer, $4/5/5/3/3/5/5/5 = 35/40$ (Cheng et al., 2026). Neither is authorized here. They remain analysis-only successors so that a higher-scoring idea cannot silently bypass the already frozen causal dependency.

3. Mechanism and Plausibility

The scanner reconstructs boundaries from the packed token tensor on every training microbatch: it compares BOS locations, performs data-dependent compaction, and materializes cumulative sequence lengths on CUDA. The packer already knows every row start and interior document start while it writes the batch. The treatment records those flattened offsets into a three-slot pinned-CPU int32 ring, asynchronously copies only the logical prefix to CUDA, and protects reuse with producer and consumer events. The same offset tensor supplies FlashAttention-3’s `cu_seqlens_q` and `cu_seqlens_k`.

The causal path is therefore

- packer-known starts → exact pinned sidecar
 - remove CUDA boundary scan
 - lower charged step time
 - more updates in 300 seconds
 - possibly lower endpoint BPB.

The mechanism is plausible only if every semantic and state mediator is identical and the timing gain clears a robust break-even. The FA3 source supports the external cumulative-boundary interface (Dao et al., 2024); it does not support the local speed or BPB claim. That claim must come entirely from the governed local contrast.

4. Frozen Phased Diagnostic

Symmetric pre-clock assay. Both registered arms independently instantiate scanner and sidecar loaders and compare exactly 1,000 batches before the diagnostic clock. Inputs, targets, epochs, flattened cumulative boundaries, boundary counts, maximum segment length, and SHA-256 digests must match. Required invariants are zero first offset, terminal offset BT , strict increase, every row start, and no gap greater than T . An interior-BOS mismatch is a hard kill.

Steps 0–19: A/A attribution prefix. Both physical arms consume scanner-derived masks. The treatment advances and safely discards its sidecar lease so producer behavior remains exercised; this conservative asymmetry is disclosed and is not used as a speed sample. Immediately after update 20, exact data, boundary, micro-loss, model, optimizer, CPU-RNG, and CUDA-RNG hashes must agree across arms.

Steps 20–39: A/B mechanism phase. At step 20, the treatment alone switches its consumed mask metadata to the sidecar; the control remains on the scanner. After update 40, the same phase-local data, boundary, loss, state, and RNG hashes must agree. Any state divergence kills the mechanism, even if timing appears favorable.

Steps 40–49: washout. These ten steps are executed but excluded from the performance estimator. They prevent the activation boundary, Python allocation transients, and phase-local instrumentation from entering the clean window.

Steps 50–249: clean profile. Exactly 200 positive finite step-time observations are retained. They form 20 consecutive blocks of 10, and the same complete blocks determine both the point estimate and its one-sided 95% lower confidence bound. The bootstrap uses the frozen seed 19019063 and 10,000 resamples. No in-clock state hashing, boundary hashing, or graph probing is admitted to this window unless charged symmetrically.

Compiler identity. The first model call may create exactly one PyTorch Dynamo graph at site 0:0. Every later model call executes under `fail_on_recompile`. Callback-derived compile IDs are hashed; zero recompiles and zero CUDA-graph recordings are required. Unsupported model or optimizer state types fail closed rather than falling back to representation-dependent hashing.

5. Two-Level Performance Decision

For the first physical placement, the treatment/control token-rate point ratio must be at least 1.054 and its one-sided 95% lower bound at least 1.041. The point threshold rounds up the robust scientific break-even from paper 019,

$$r_{\text{robust}} = \exp(0.002396/0.046) = 1.05347.$$

The separate 1.041 lower-bound screen approximates the central, rather than robust, token-law break-even and is a fixed diagnostic decision bound rather than population-level experimental confidence. Failure of either preregistered diagnostic threshold kills Round 1 before validation scoring.

Passing both thresholds does not authorize a 300-second endpoint. One GPU may be systematically faster than the other, and temporal bootstrapping cannot identify placement bias. A pass yields only REQUIRE_COUNTERBALANCED_PROFILE. A separately frozen role-swapped AB/BA profile must reuse the same source, challenge, UUID bindings, phase contract, and complete-block estimator. Only a counter-balanced pass may promote the ordinary two-arm endpoint proposal through the normal typed gate. Thus a single placement can kill or request stronger evidence, but cannot create a favorable BPB opportunity.

Each pair artifact must bind exactly two distinct immutable RunRecords, their fingerprints, roles, run UUIDs, source snapshot, challenge fingerprint, and resolved configurations. The sole resolved arm difference is PACKER_DOC_BOUNDARIES. A child failure still writes an INVALID_DIAGNOSTIC pair artifact. Recovery may finalize comparison from existing records but may never relaunch the same immutable diagnostic.

6. H200 Systems Evidence Before Execution

Two independent non-experimental validators ran on H200s: one with ordinary asynchronous CUDA execution and one with CUDA_LAUNCH_BLOCKING=1. Both produced the same terminal digest and passed delayed H2D, delayed consumer work across ring wraps, outstanding fourth-lease rejection, wrong-stream rejection, discard contracts, poisoned tail lengths, exact capacity bounds, a single dynamic compile identity, forced static-shape recompile rejection, and real FA3 varlen forward/backward bitwise equivalence over nine ring-wrap iterations. The FA3 check used a synthetic 32-token, two-head, head-dimension-64 workload and a small compiled helper. It did not

Table 2. Preflight evidence. “Pass” is systems scope only.

check	result
runtime	PyTorch 2.9.1+cu128; CUDA 12.8
device	NVIDIA H200, SM90
event and ring lifetime	pass
Dynamo graphs / recompiles	1 / 0
forced recompile	rejected
real FA3 forward/backward	bitwise exact
async vs blocking log SHA	identical

execute the full training graph.

Two predecessor attempts are part of the record. The first failed before GPU work because the standalone validator lacked the repository root on sys.path; the second found that applying fail_on_recompile to the initial call rejects the required first compile. The frozen implementation now permits the first compile and applies the stance to all later calls. These failures improve attribution; they are not discarded failed trials.

The validated FA3 result digest is e5a96b00475fbcf1425eb72315ef0ef02e6138fc82cb84983529ef452f750127. The two log files share SHA-256 37bae79b44198818fce02ef54a6a26482f119505511a547fa9fd6a17759ac906. This evidence is neither a RunRecord nor a throughput measurement.

7. Retrospective Round-2 Update

Paper 019’s canonical-key Round 2 was independently evaluated by its frozen offline mediator before any GPU scoring. The 8,192-token vocabulary reduced to 6,808 canonical keys (16.8945%), while the conservative observed affected shares were 17.7887% for bigrams and 3.7155% for trigrams. The governing minimum, 3.7155%, was below the preregistered 10% gate. The result is

KILL_BEFORE_GPU_SCORING.

The result and output-manifest SHA-256 values are 9d0e3a122a7e880471bc793368576a89afa5cfbe442f1149c1bbc651eed4d761 and d15dc6b3be202b463befd270abf1af04b8b24c3ab3a6acce831a7a5a58189f63. Pooled or less conservative shares cannot replace the frozen statistic. This result was known only after paper 019 and may update the portfolio, but it may not tune Round-1 thresholds. A future paper may replace the dead round with the independently ranked prime-modulus $\rightarrow$ constant-parameter $K = 4 \rightarrow$ contextual-gate Engram chain.

8. Reproducibility Binding

The exact systems-validation manifest is stored under `research/experiments/artifacts/paper019_round1_fa3_boundary_sidecar`. At this addendum freeze, the principal source SHA-256 values are:

```
lib.py
13de28bcb398866be65cfca489413818d75dc2fcd2461e367be06198f
879123d
train.py
e2d4ac9e91c6ef716a8fdab35ab72da3e0e4bb23cbbecee04cc528b66
d172b8b
tools/run_gated.py
47d08ec38f6dff53ed50cdc5e4ddaf88342da0f2e9c5993219baf41217
11868a
tools/run_stage.py
c7d3ab45af2983598936859ea9f14c34ebb6da3ce9b1d0cc9d00ca6c
764b16c0
tools/register_paper019_fa3_boundary_sidecar_chain.py
d8f6b4e71a65c971ceffdc63630b99daa4d4cd57c3546cc52410558e
e8e9ce0
tools/validate_packer_sidecar_h200.py
e5d58f8d0a3db1a87b1e82ee1b4063da92e49eac015a7c6b6e0f7203
79a1c9e4
H200 validation manifest
f0a226c34198186bfce9a047e328d2b4e3fb290a3993dc1cb24829792
b37c213
```

Any later edit invalidates these bindings and requires a new reconciliation and explicit amendment before execution. Remote launch must independently match the local source hashes and the pinned FA3 repository.

9. Conclusion

This addendum converts an attractive exact-semantics optimization into a failure-attributable sequence. It separates semantic equivalence, state equivalence, compiler identity, systems lifetime, clean timing, placement bias, and endpoint BPB. The candidate can advance only by clearing each link in order; a favorable downstream number cannot repair an upstream failure. That discipline is the mechanism by which exploration remains novel without becoming ungrounded.

References

- GPT Research Coordinator. Observe, Attribute, Then Improve: A Critic-Ranked Five-Mechanism Program for an Autoresearch Plateau. OPHIS paper 019, 2026.
- T. Dao et al. FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision. NeurIPS, 2024.
- T. Chen et al. GPAS: Accelerating Convergence of LLM Pretraining via Gradient-Preserving Activation Scaling. NeurIPS, 2025.
- X. Cheng et al. Conditional Memory via Scalable Lookup: A New Axis of Sparsity for Large Language Models. ACL, 2026.